Chapter III.3

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3.1. Let X be a noetherian scheme. Show that X is affine if and only if X_{red} is affine. [Hint: Use (3.7), and for any coherent sheaf $\mathscr{F}$ on X, consider the filtration $\mathscr{F} \supset \mathscr{N} \cdot \mathscr{F} \supset \mathscr{N}^2 \cdot \mathscr{F} \supset \cdots$, where $\mathscr{N}$ is the sheaf of nilpotent elements on X.]

Proof. If $X = \operatorname{Spec} A$ is affine, then $X_{\text{red}} = \operatorname{Spec}(A/\operatorname{nil} A)$ is affine. Conversely, suppose that X_{red} is affine. For any quasi-coherent sheaf $\mathscr{F}$ on X, consider the filtration $\mathscr{F} \supset \mathscr{N}\mathscr{F} \supset \mathscr{N}^2\mathscr{F} \supset \cdots$, where $\mathscr{N}$ is the sheaf of nilpotent elements on X.

Noetherianity of X gives us a finite affine open cover $\{U_i\}$, where each $\mathscr{N}(U_i)$ is a finitely generated $\mathscr{O}_X(U_i)$-module since $\mathscr{O}_X(U_i)$ is noetherian. Thus, we can take $n \geqslant 0$ such that $\mathscr{N}^n(U_i) = 0$ for each i, whence $\mathscr{N}^n = 0$. This gives us a finite filtration

$$0 = \mathscr{N}^n\mathscr{F} \subset \cdots \subset \mathscr{N}\mathscr{F} \subset \mathscr{F},$$

and thus the short exact sequences

$$0 \to \mathscr{N}^{k+1}\mathscr{F} \to \mathscr{N}^k\mathscr{F} \to \frac{\mathscr{N}^k\mathscr{F}}{\mathscr{N}^{k+1}\mathscr{F}} \to 0 \tag{1}$$

for all $0 \leqslant k < n$.

Clearly

$$\mathscr{N} \subset \operatorname{Ann}\left(\frac{\mathscr{N}^k\mathscr{F}}{\mathscr{N}^{k+1}\mathscr{F}}\right),$$

so there is a quasi-coherent scheme $\mathscr{G}$ on X_{red} such that

$$\frac{\mathscr{N}^k\mathscr{F}}{\mathscr{N}^{k+1}\mathscr{F}} \cong i_*\mathscr{G}$$

with $i : X_{\text{red}} \to X$ the inclusion. It follows from (2.10) that

$$H^\bullet(X, \frac{\mathscr{N}^k\mathscr{F}}{\mathscr{N}^{k+1}\mathscr{F}}) \cong H^\bullet(X_{\text{red}}, \mathscr{G}).$$

(3.7) implies $H^i(X_{\text{red}}, \mathscr{G}) = 0$ for $i > 0$, so the long exact sequence of homology for (1) gives us exactness of

$$H^i(X, \mathscr{N}^{k+1}\mathscr{F}) \to H^i(X, \mathscr{N}^k\mathscr{F}) \to H^i(X_{\text{red}}, \mathscr{G}) = 0$$

for all $i > 0$. $H^\bullet(X, \mathscr{N}^n\mathscr{F}) = 0$, so induction gives us $H^i(X, \mathscr{N}^k\mathscr{F}) = 0$ for all $i > 0$ and $0 \leqslant k \leqslant n$. In particular, $H^i(X, \mathscr{F}) = 0$ for all $i > 0$. $\mathscr{F}$ is an arbitrary quasi-coherent sheaf on X, so that X is affine by (3.7). □

3.2. Let X be a reduced noetherian scheme. Show that X is affine if and only if each irreducible component is affine.

Proof. One implication is clear. Suppose, then, that each irreducible component of X is affine. X is noetherian, so we can write a finite irreducible decomposition $X = Y_1 \cup \cdots \cup Y_r$, with each $Y_1, \ldots, Y_r$ affine. By (Ex. 3.1), we may assume X is reduced. Now, fix a quasi-coherent sheaf $\mathscr{F}$ on X. We have a filtration

$$\mathscr{F} \supset \mathscr{I}_{Y_1}\mathscr{F} \supset \mathscr{I}_{Y_1 \cup Y_2}\mathscr{F} \supset \cdots \supset \mathscr{I}_{Y_1 \cup \cdots \cup Y_r}\mathscr{F} = \mathscr{I}_X = 0,$$

where each $Y_1 \cup \cdots \cup Y_k$ is given the reduced induced subscheme structure. This gives us exact sequences

$$0 \to \mathscr{I}_{Y_1 \cup \cdots \cup Y_{k+1}}\mathscr{F} \to \mathscr{I}_{Y_1 \cup \cdots \cup Y_k}\mathscr{F} \to \frac{\mathscr{I}_{Y_1 \cup \cdots \cup Y_k}\mathscr{F}}{\mathscr{I}_{Y_1 \cup \cdots \cup Y_{k+1}}\mathscr{F}} \to 0$$

for all $0 \leqslant k < r$, where

$$\mathscr{I}_{Y_{k+1}} \subset \operatorname{Ann}\left(\frac{\mathscr{I}_{Y_1 \cup \cdots \cup Y_k}\mathscr{F}}{\mathscr{I}_{Y_1 \cup \cdots \cup Y_{k+1}}\mathscr{F}}\right).$$

Then by (2.10) we have quasi-coherent sheaves $\mathscr{G}_{k+1}$ on Y_{k+1} such that

$$H^\bullet(X, \frac{\mathscr{I}_{Y_1 \cup \cdots \cup Y_k}\mathscr{F}}{\mathscr{I}_{Y_1 \cup \cdots \cup Y_{k+1}}\mathscr{F}}) \cong H^\bullet(Y_{k+1}, \mathscr{G}_{k+1}).$$

each Y_{k+1} is affine, so $H^p(Y_{k+1}, \mathscr{G}_{k+1}) = 0$ for all $p > 0$. Hence, the maps

$$H^p(X, \mathscr{I}_{Y_1 \cup \cdots \cup Y_{k+1}}\mathscr{F}) \to H^p(X, \mathscr{I}_{Y_1 \cup \cdots \cup Y_k}\mathscr{F})$$

are surjective for all $0 \leqslant k < r$ and $p > 0$ as in (Ex. 3.1). It follows that $H^p(X, \mathscr{F}) = 0$ for all $p > 0$ by induction on $r - k$, so that X is affine by (3.7). □

3.3. Let A be a noetherian ring, and let $\mathfrak{a}$ be an ideal of X.

(a) Show that $\Gamma_\mathfrak{a}(\cdot)$ (II, Ex. 5.6) is a left-exact functor from the category of A-modules to itself. We denote its right derived functors, calculated in $\mathfrak{Mod}(A)$, by $H^i_\mathfrak{a}(\cdot)$.

(b) Now let $X = \operatorname{Spec} A$, $Y = V(\mathfrak{a})$. Show that for any A-module M,

$$H^i_\mathfrak{a}(M) = H^i_Y(X, \widetilde{M}).$$

(c) For any i, show that $\Gamma_\mathfrak{a}(H^i_\mathfrak{a}(M)) = H^i_\mathfrak{a}(M)$.

Proof. (a) Let $0 \to M' \xrightarrow{\phi} M \xrightarrow{\psi} M''$ be a left exact sequence in $\mathfrak{Mod}_A$, and consider the following diagram.

$$\begin{array}{ccccccc} 0 & \longrightarrow & M' & \xrightarrow{\phi} & M & \xrightarrow{\psi} & M'' \\ & & \uparrow & & \uparrow & & \uparrow \\ 0 & \longrightarrow & \Gamma_\mathfrak{a}(M') & \xrightarrow{\phi_\mathfrak{a}} & \Gamma_\mathfrak{a}(M) & \xrightarrow{\psi_\mathfrak{a}} & \Gamma_\mathfrak{a}(M'') \end{array}$$

We clearly have $\phi_\mathfrak{a}$ injective and $\operatorname{im}\phi_\mathfrak{a} \subset \ker\psi_\mathfrak{a}$. It suffices to show $\ker\psi_\mathfrak{a} \subset \operatorname{im}\phi_\mathfrak{a}$.

Take $m \in \ker\psi_\mathfrak{a} \subset \ker\psi$, and $m' \in M'$ such that $\phi(m') = m$. We have $n > 0$ such that $0 = \mathfrak{a}^n\phi(m') = \phi(\mathfrak{a}^n m')$, whence $\mathfrak{a}^n m' = 0$ by injectivity of ϕ. We conclude that $m' \in \Gamma_\mathfrak{a}(M')$, so that $m = \phi_\mathfrak{a}(m') \in \operatorname{im}\phi_\mathfrak{a}$ gives us exactness.

(b) Let $0 \to M \to I^\bullet$ be an injective resolution of M in $\mathfrak{Mod}_A$. Then $\widetilde{I^\bullet}$ is a flasque resolution of $\widetilde{M}$ by (3.4). Thus, it suffices to show that $\Gamma_\mathfrak{a}(I^\bullet) \cong \Gamma_Y(\widetilde{I^\bullet})$. This follows from the more general fact that $\Gamma_\mathfrak{a}(N) \cong \Gamma_Y(\widetilde{N})$, which is a corollary of (II, Ex. 5.6c).

(c) Let $0 \to M \to I^\bullet$ be an injective resolution of M, so that $H^\bullet_{\mathfrak{a}}(M) = h^\bullet(\Gamma_{\mathfrak{a}}(I^\bullet))$. Then, with boundary maps $d^p : \Gamma_{\mathfrak{a}}(I^p) \to \Gamma_{\mathfrak{a}}(I^{p+1})$,

$$\begin{aligned}\Gamma_{\mathfrak{a}}(h^p(\Gamma_{\mathfrak{a}}(I^\bullet))) &= \Gamma_{\mathfrak{a}}\left(\frac{\ker d^p}{\operatorname{im} d^{p-1}}\right) \\ &\supset \frac{\Gamma_{\mathfrak{a}}(\ker d^p)}{\operatorname{im} d^{p-1}} \\ &= h^p(\Gamma_{\mathfrak{a}}(I^\bullet)),\end{aligned}$$

since $\ker d^p \subset \Gamma_{\mathfrak{a}}(I^p)$. That is,

$$\Gamma_{\mathfrak{a}}(H^\bullet_{\mathfrak{a}}(M)) = H^\bullet_{\mathfrak{a}}(M),$$

as desired. □

3.7 Let A be a noetherian ring, let $X = \operatorname{Spec} A$, let $\mathfrak{a} \trianglelefteqslant A$ be an ideal, and let $U \subset X$ be the open set $X - V(\mathfrak{a})$.

(a) For any A-module M, establish the following formula of Deligne:

$$\Gamma(U, \widetilde{M}) \cong \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M).$$

(b) Apply this in the case of an injective A-module I to give another proof of (3.4).

Proof. (a) Fix $f \in \mathfrak{a}$, and consider the direct system $\{\operatorname{Hom}_A(\mathfrak{a}^n, M)\}_{n>0}$ with compatibility given by restriction. For any $n > 0$, we have a natural map

$$\begin{aligned}\Psi_{n,f} : \operatorname{Hom}_A(\mathfrak{a}^n, M) &\longrightarrow M_f \\ \phi &\longmapsto \frac{\phi(f^n)}{f^n}.\end{aligned}$$

We see that

$$\begin{aligned}\Psi_{n+1,f}(\phi|_{\mathfrak{a}^{n+1}}) &= \frac{\phi|_{\mathfrak{a}^{n+1}}(f^{n+1})}{f^{n+1}} \\ &= \frac{\phi(f^n)}{f^n} \\ &= \Psi_{n,f}(\phi),\end{aligned}$$

so the $\Psi_{n,f}$ induce a natural map

$$\Psi_f : \varinjlim_n \operatorname{Hom}_A(\mathfrak{a}^n, M) \to M_f.$$

Surjectivity is clear, as $\Psi_f(f^n \mapsto m) = m/f^n \in M_f$ for arbitrary $m \in M$ and $n > 0$.

Recall that

$$\Gamma(U, \widetilde{M}) = \varprojlim_{f \in \mathfrak{a}} \Gamma(D_f, \widetilde{M}) = \varprojlim_{f \in \mathfrak{a}} M_f,$$

where the inverse system is given by $g \leqslant f$ iff $\sqrt{(g)} \subset \sqrt{(f)}$, with maps

$$\begin{aligned}\rho^f_g : M_f &\longrightarrow M_g \\ \frac{m}{f^n} &\longrightarrow \frac{mu^n}{g^{kn}},\end{aligned}$$

where $u \in A$ such that $g^k = uf$ is supplied by $(g) \subset \sqrt{(g)} \subset \sqrt{(f)}$. We compute

$$\begin{aligned}\rho_g^f \Psi_f(\phi) &= \rho_g^f \left(\frac{\phi(f^n)}{f^n} \right) \\ &= \frac{\phi(f^n)u^n}{g^{kn}} \\ &= \frac{\phi(f^n u^n)}{g^{kn}} \\ &= \frac{\phi(g^{kn})}{g^{kn}} \\ &= \Psi_g(\phi),\end{aligned}$$

so that the Ψ_f glue to give us a natural morphism $\Psi : \varinjlim \mathrm{Hom}_A(\mathfrak{a}^n, M) \to \Gamma(U, \widetilde{M})$. It suffices to show that this map is indeed an isomorphism.

We begin by showing the map is injective. If $\Psi(\phi) = 0$, then $\Psi(\phi)|_{D_f} = 0$ for all $f \in \mathfrak{a}$. Thus, for any $f \in \mathfrak{a}$ there exists $n > 0$ such that $\phi(f^n)/f^n = 0 \in M_f$. That is,

$$\phi(f^{n+k}) = f^k \phi(f^n) = 0$$

for $k \geqslant 0$. Taking generators $f_1, \ldots, f_r \in \mathfrak{a}$ supplied by noetherianity of A, we have that

$$\phi(\mathfrak{a}^n) = \phi((f_1, \ldots, f_r)^n) = 0$$

for $n \gg 0$. Hence, $\phi = 0 \in \varinjlim \mathrm{Hom}_A(\mathfrak{a}^n, M)$.

We conclude by showing that Ψ is surjective. Fix $s \in \Gamma(U, \widetilde{M})$, and recall that our generators $f_1, \ldots, f_r$ give us a finite cover $\{D_{f_i}\}_{i=1}^r$ of U. It suffices to find $\phi \in \varinjlim \mathrm{Hom}_A(\mathfrak{a}^n, M)$ such that $\Psi_{f_i}(\phi) = s_i := s|_{D_{f_i}}$ for each $i = 1, \ldots, r$. Recall that Ψ_{f_i} is surjective, so we can choose ϕ_i such that $\Psi_{f_i}(\phi_i) = s_i$. Then $s_i = \phi_i(f_i^n)/f_i^n$ for $n \gg 0$, so

$$\frac{\phi_i(f_i^n) f_j^n}{f_i^n f_j^n} = s_i|_{D_{f_j}} = s_j|_{D_{f_i}} = \frac{f_i^n \phi_j(f_j^n)}{f_i^n f_j^n}$$

for i, j. Thus,

$$\phi_i(f_i^n f_j^n) = \phi_j(f_i^n f_j^n) \in M$$

for $n \gg 0$.

We define a map $\phi : \mathfrak{a}^n \to M$ for $n \gg 0$ on the generators $\{f_1^{e_1} \cdots f_r^{e_r}\}_{\sum_i e_i = n}$ of $\mathfrak{a}^n = (f_1, \ldots, f_r)^n$ by

$$\phi(f_1^{e_1} \cdots f_r^{e_r}) = \phi_i(f_1^{e_1} \cdots f_r^{e_r}),$$

where i maximizes e_i. If $e_i = e_j$ for some i, j, then $e_i, e_j \geqslant n/r \gg 0$, so that

$$\phi_i(f_1^{e_1} \cdots f_r^{e_r}) = \phi_j(f_1^{e_1} \cdots f_r^{e_r})$$

by the above. If we can write a generator as $f_1^{e_1} \cdots f_r^{e_r} = a f_1^{l_1} \cdots f_r^{l_r}$, where $e_i \neq l_j$ and $a \in A$, then we can remove the former term from our generating set. It suffices, then, to show that ϕ is an A-module homomorphism.

Fix some $n > 0$ so that there are well-defined representatives $\phi_i : \mathfrak{a}^n \to M$ for each $i = 1, \ldots, r$, and $\phi_i(f_i^n f_j^n) = \phi_j(f_i^n f_j^n)$. We define a morphism $\phi : (f_1^n, \ldots, f_r^n) \to M$ by $\phi(f_i^n) = \phi_i(f_i^n)$.

(b) Fix an injective A-module I, let $V \subset U$ be another open set, and let $\mathfrak{b} \trianglelefteq A$ an ideal such that $V = X - V(\mathfrak{b})$. Then $V(\mathfrak{a}) \subset V(\mathfrak{b})$, so $\sqrt{\mathfrak{a}} \supset \sqrt{\mathfrak{b}}$. A noetherian implies $\mathfrak{a}, \mathfrak{b}$

finitely generated, so we can find $m > 0$ such that $\mathfrak{a} = \sqrt{\mathfrak{a}}^m$ and $\mathfrak{b} = \sqrt{\mathfrak{b}}^m$. Then $\mathfrak{a} \supset \mathfrak{b}$, so I injective implies the restriction map

$$\mathrm{Hom}_A(\mathfrak{a}^n, I) \to \mathrm{Hom}_A(\mathfrak{b}^n, I)$$

is surjective for any n. A direct limit of epimorphisms is itself an epimorphism, so

$$\Gamma(U, \widetilde{I}) \to \Gamma(V, \widetilde{I})$$

is surjective by (a). $\square$